Imperial College London

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Advanced Imaging Technologies in Systems Biology and Biomedical Research

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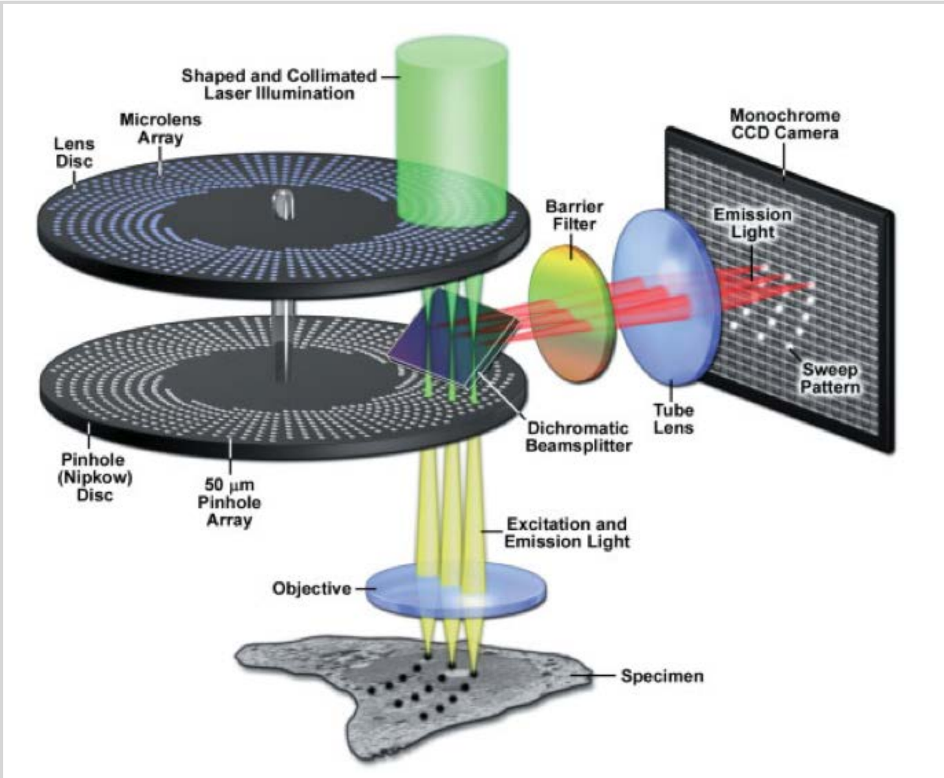
What we covered

- How to perform fast imaging?
- How to perform fast volumetric imaging?
- How to visualise molecular dynamics using fluorescence imaging I?

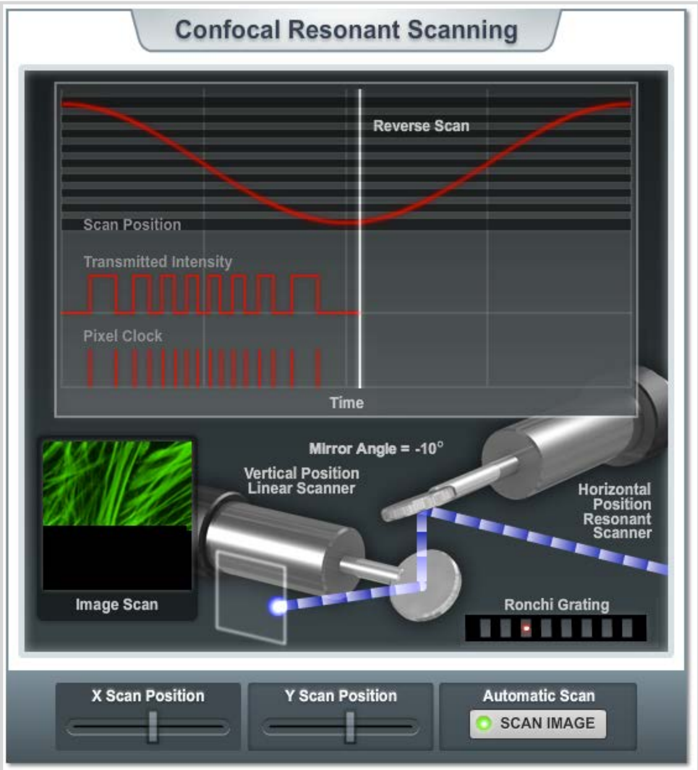
Confocal Imaging with a Spinning Disk: The Yokogawa design

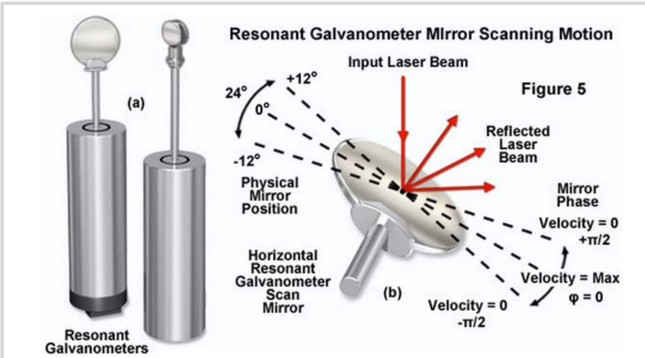
70% incident ligth delivered
1-2%fluorescent emission captured



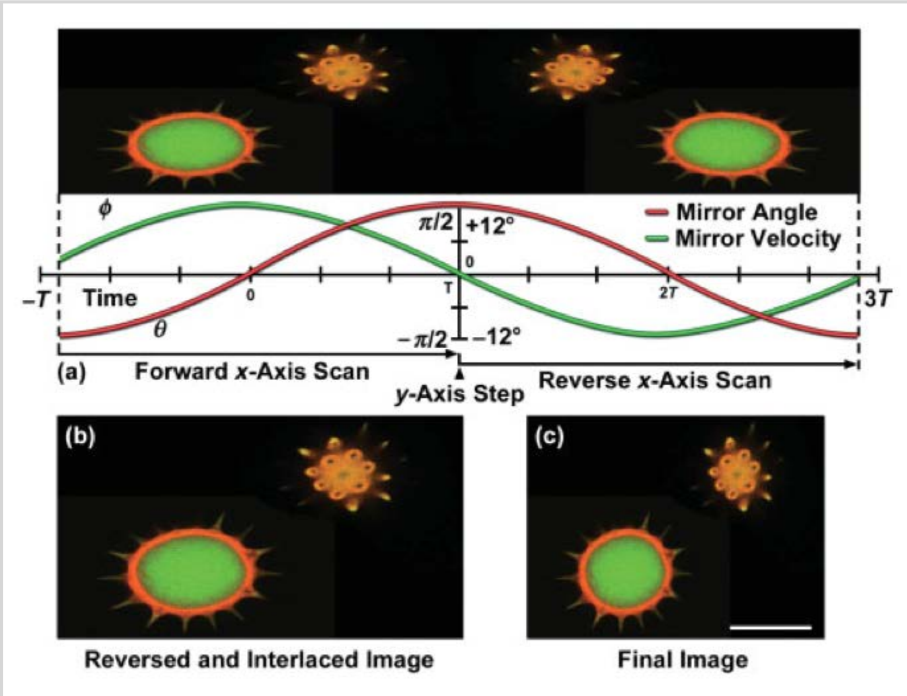
The pinhole spiral patterns are designed so that a complete image is created with each 30° rotation of the disk.

Resonant Scanning in Laser Scanning Confocal Microscopy

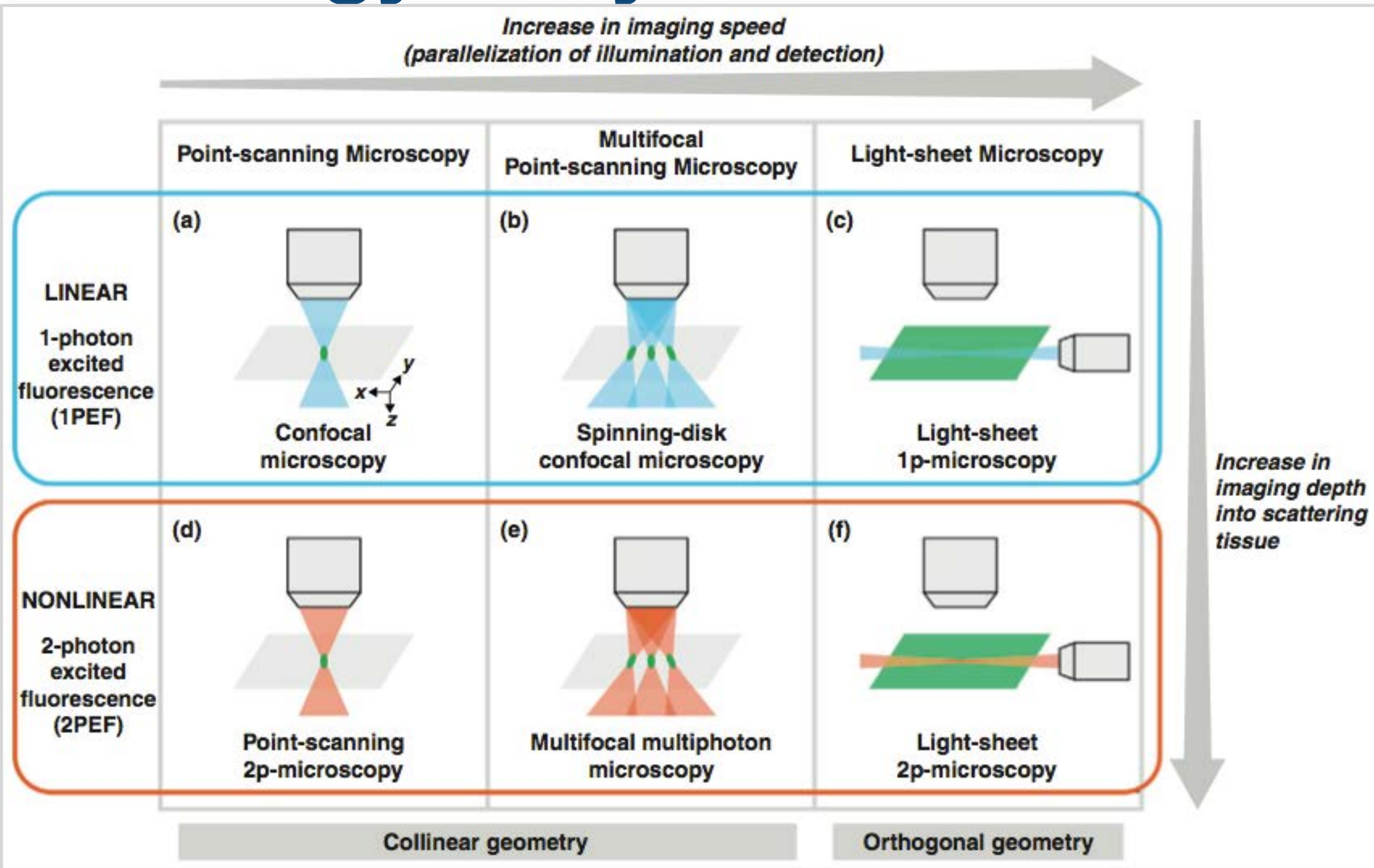




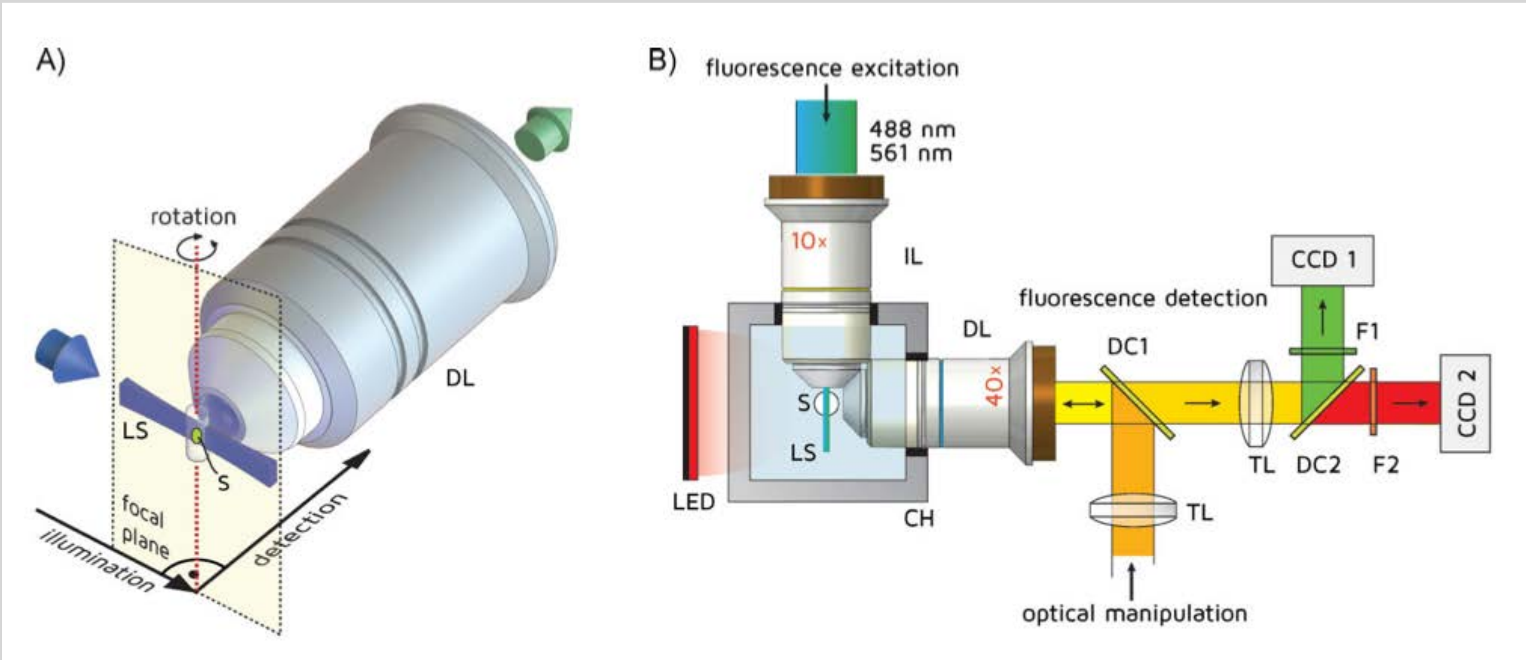
Resonant galvanometers display a nonlinear velocity



Why the need to image fast in space? Biology is dynamic and 3D!



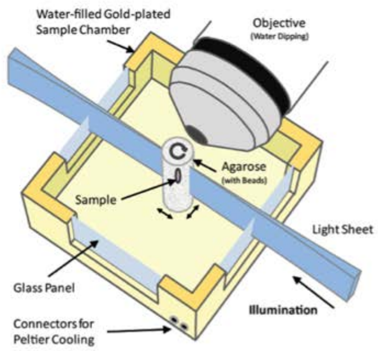
Selective plane illumination microscopy (SPIM) concept and setup



Only a thin slice of the sample is illuminated by the light sheet and imaged onto the camera; no out-of-focus blur is created.
Photo-toxicity and bleaching of the fluorophores are reduced to a minimum.

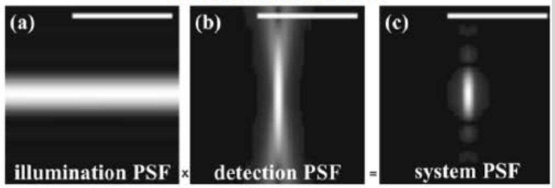
Resolution of light sheet microscopy

Resolution of light sheet microscopy



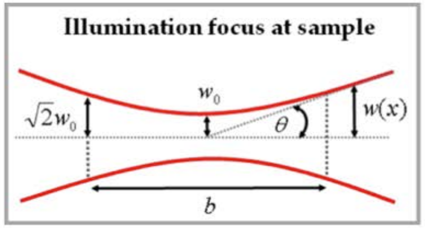
- **Lateral resolution:**
  - ➔ **detection lens**
- **Axial resolution:**
  - ➔ **illumination and detection lens**
- **Field of view:**
  - ➔ **confocal parameter of illumination**

Axial resolution

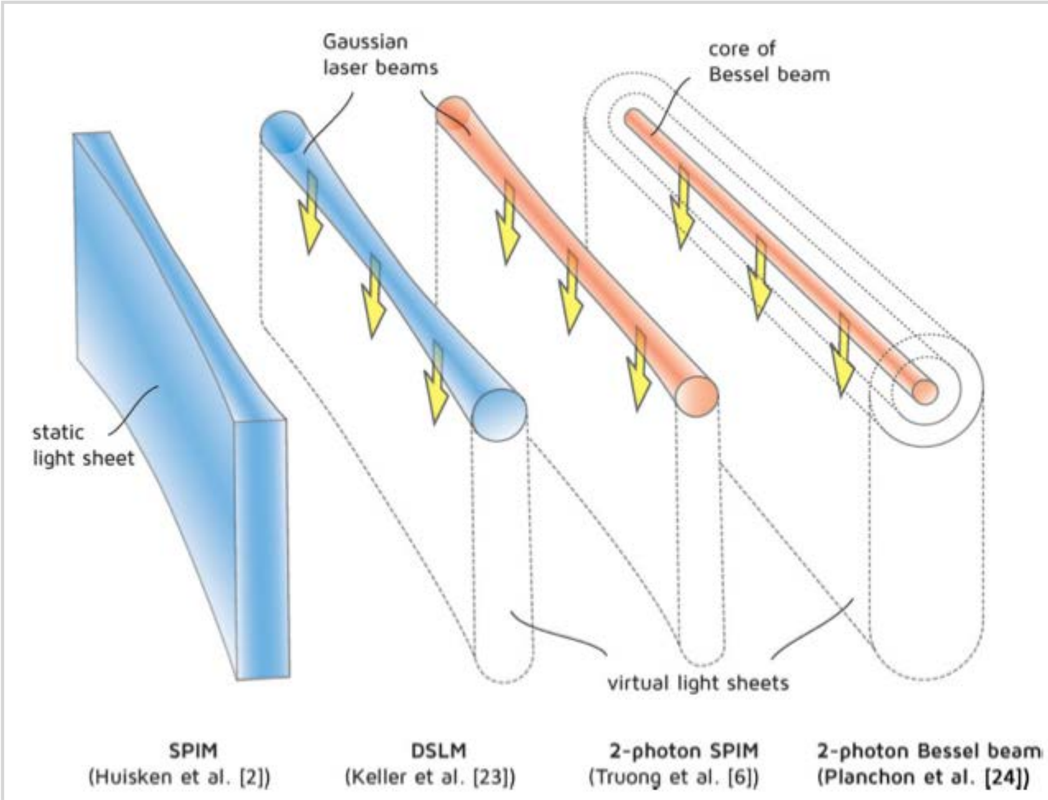


Sheet thickness : $w_0 = \lambda / (\pi \cdot NA_{ill})$

Field of view : $b = 2n\lambda / (\pi \cdot NA_{ill}^2)$



Different forms of light sheets



Modes of dynamic fluorescence imaging

FRAP Fluorescence recovery after photobleaching.
FLAP Fluorescence localisation after photobleaching.
FLIP Fluorescence loss in photobleaching.
FDAP Fluorescence decay after photoactivation.

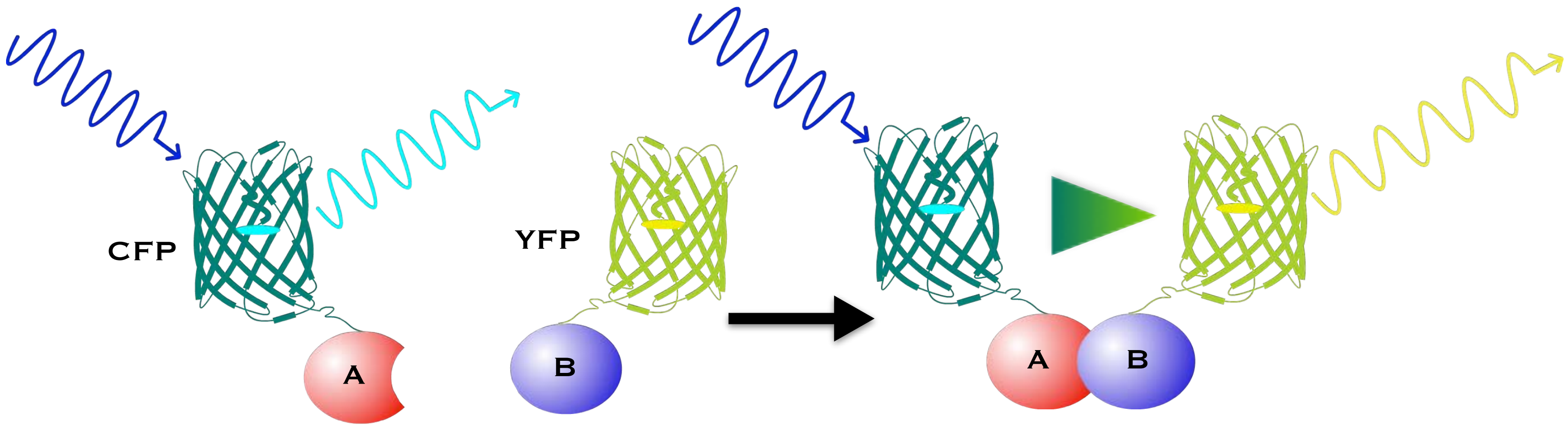
4D/5D

Time lapse microscopy
in x, y, and z
spatial dimensions in time (4D imaging)
and in multiple colour channels (5D imaging).

FCS Fluorescence correlation spectroscopy.
FCCS Fluorescence cross-correlation spectroscopy.
ICS Image correlation spectroscopy.

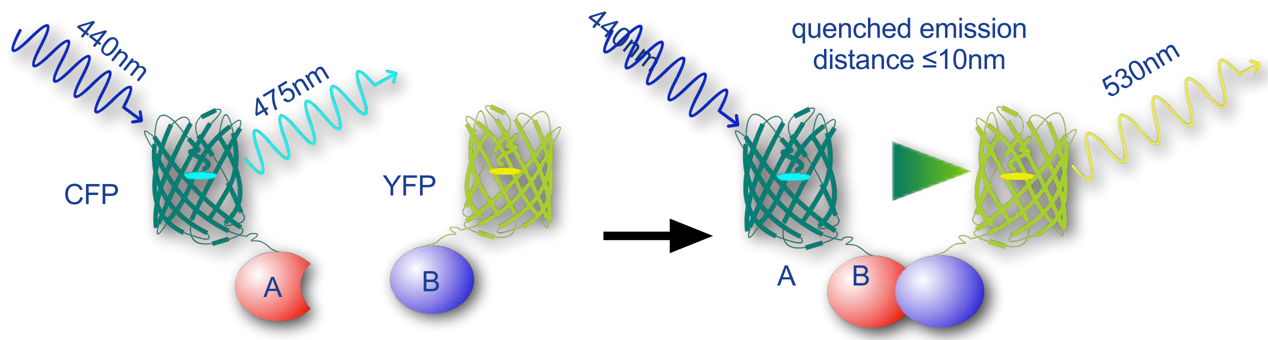
FRET Förster (fluorescence) resonance energy transfer.
BRET Bioluminescence resonance energy transfer.
BIFC Bimolecular Bimolecular fluorescence complementation.
FLIM Fluorescence lifetime imaging microscopy.

Molecular mechanism of FRET



FRET involves the radiation-less transfer of energy between closely spaced donor and acceptor fluorophores.
FRET occurs in the absence of emission and absorption of an intermediate photon.

Main requirements for FRET



- Fluorophore proximity (up to 10nm)
(Förster distance R0 = 50% FRET transfer probability)

- Dipole alignment
(is greatest when both dipoles are in line and parallel and least when they are perpendicular to one another)

- Spectral overlap
(emission spectrum of the donor fluorophore overlap the excitation spectrum of the acceptor fluorophore)

- Environmental conditions
(affect dipole interactions, including solvent type, degree of hydration, ionic strength, pH, and temperature)

Evaluating FRET from fluorescent images

one can calculate the FRET efficiency, E, and the intermolecular distance, R.

E is calculated from the ratio of fluorescence intensities
of the donor with acceptor (I_{da}) and without the acceptor (I_d):

$$E = 1-(I_{da} / I_d).$$

FRET efficiency is related to the intermolecular distance R between the donor and acceptor fluorophores.

Since E and R are related:

$$E=R_0^6/(R_0^6 + R^6)$$

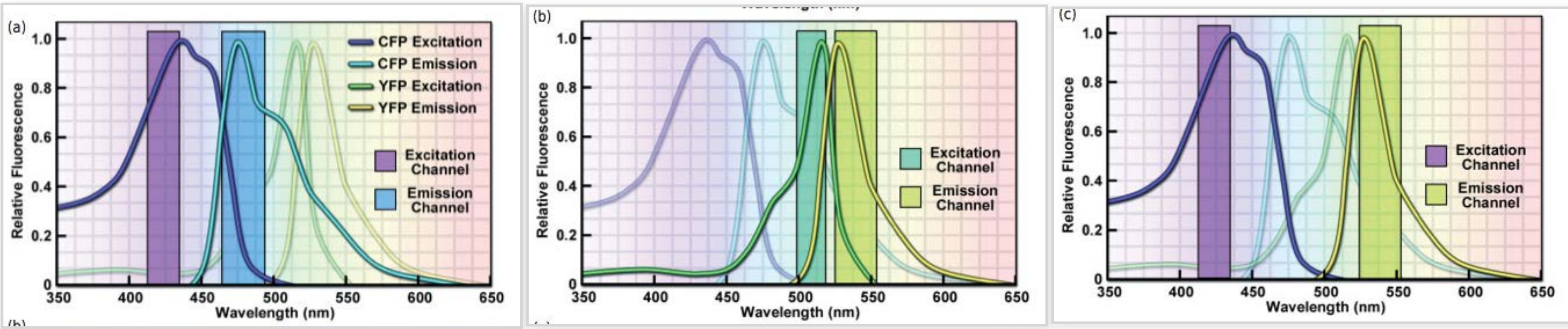
dipole–dipole coupling and FRET efficiency are related to the inverse 6th power of the donor–acceptor distance. This degree of sensitivity makes FRET very useful as a “spectroscopic ruler” for measuring intermolecular distances in the range of 1–10 nm.

Paradigms for FRET

• **Intermolecular FRET:**
Donor–acceptor fluorophores are located on two separate molecules that interact with each other, and FRET indicates the proximity of the two reporters, and indirectly, the proximity of the two test molecules.

• **Intramolecular FRET:**
Donor-acceptor fluorophores are placed on different regions of a single polypeptide that undergoes conformational changes upon binding to a substrate, causing the fluorophores involved in FRET to come together or move apart.

Methods for detecting FRET: Sensitised Acceptor Emission FRET



The operator must calculate correction factors for:

- **donor bleedthrough** (donor fluorescence that contributes to signal in the FRET image)
- **crosstalk** (donor excitation that directly excites acceptor fluorescence).

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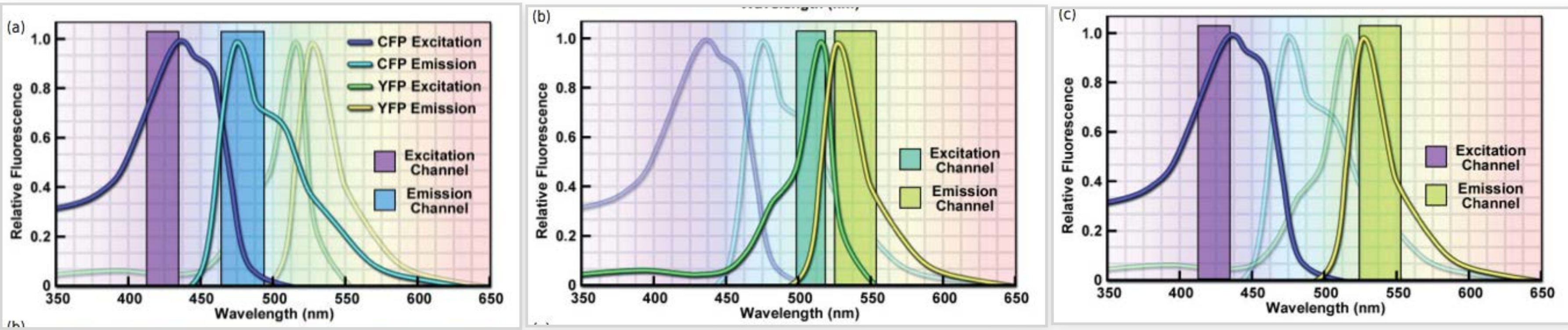

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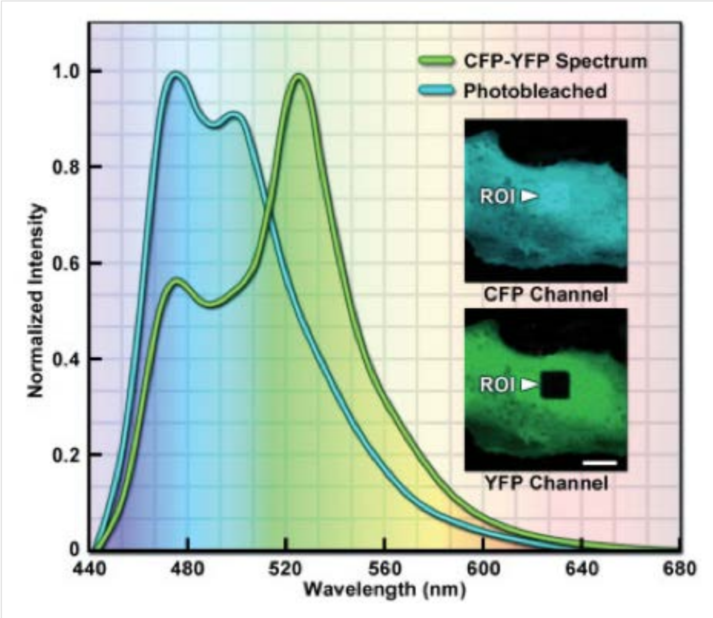
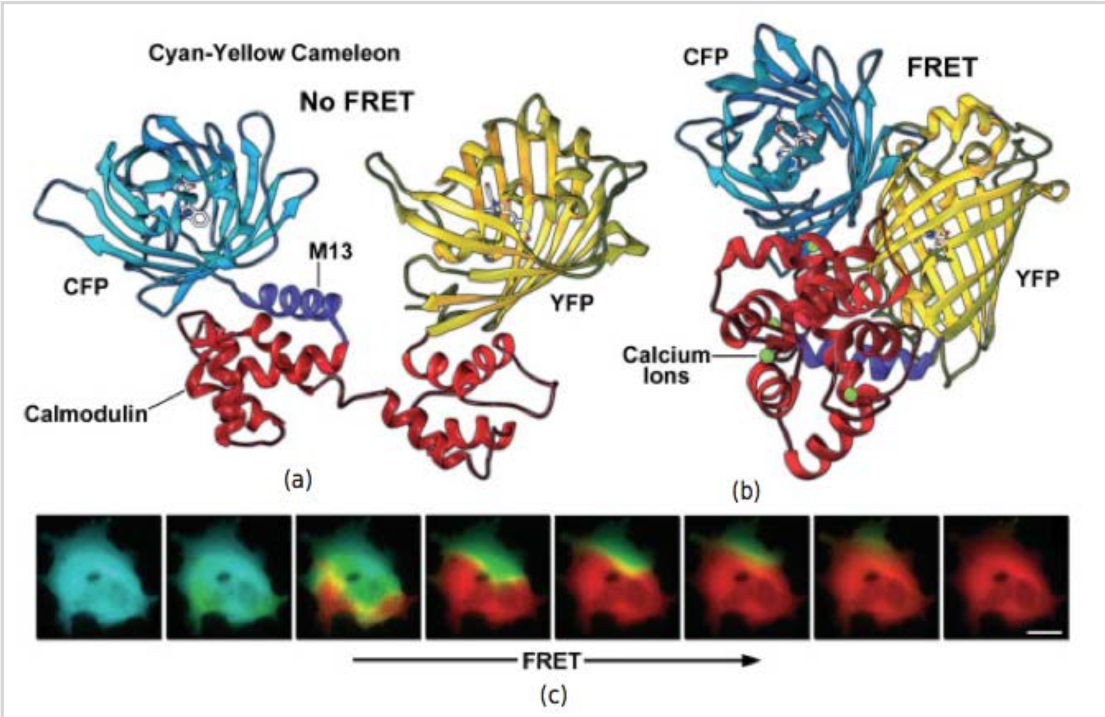
Methods for detecting FRET: Acceptor-Photobleaching FRET with de-quenching of donor fluorescence



The operator must ensure that
the illumination used to bleach acceptor fluorescence does not also destroy some of the donor fluorescence.

Applications for FRET: Cameleons-biosensors for Ca^{2+} ions



FRET: additional considerations

- The concentrations of donor and acceptor need to be controlled so that the efficiency of FRET transfer is optimal. Ideally, there is molar excess of acceptor over donor (but not a 10-fold or greater molar excess).
- Photobleaching should be minimal to maximise the number of functional molecules generating the FRET signal.
- Spectra for donor emission and acceptor excitation must overlap substantially.
- The best FRET pairs have a donor with a large fluorescence quantum yield and an acceptor with a large absorption extinction coefficient.
- Minimise direct excitation of acceptor during excitation of donor.
- Donor’s absorption and emission spectra should overlap minimally to avoid donor–donor transfer, the case where emission of the donor molecules stimulates other donors to fluoresce, thus increasing a nonspecific background signal.
- Labeled donor and acceptor molecules must retain their biological functions.
- Direct excitation of the acceptor at the donor excitation wavelength. This is unwanted but is usually unavoidable, since the excitation spectra overlap and have long tails.
- Possibility of Emission of the excited donor at the acceptor emission wavelengths.
- Effects of varying concentrations of donor and acceptor molecules.
- Poor filter performance and therefore poor spectral discrimination when performing FRET.

What we will cover

- How to visualise molecular dynamics using fluorescence imaging II?
- What are the basic principles of development?
- How is tissue patterned?

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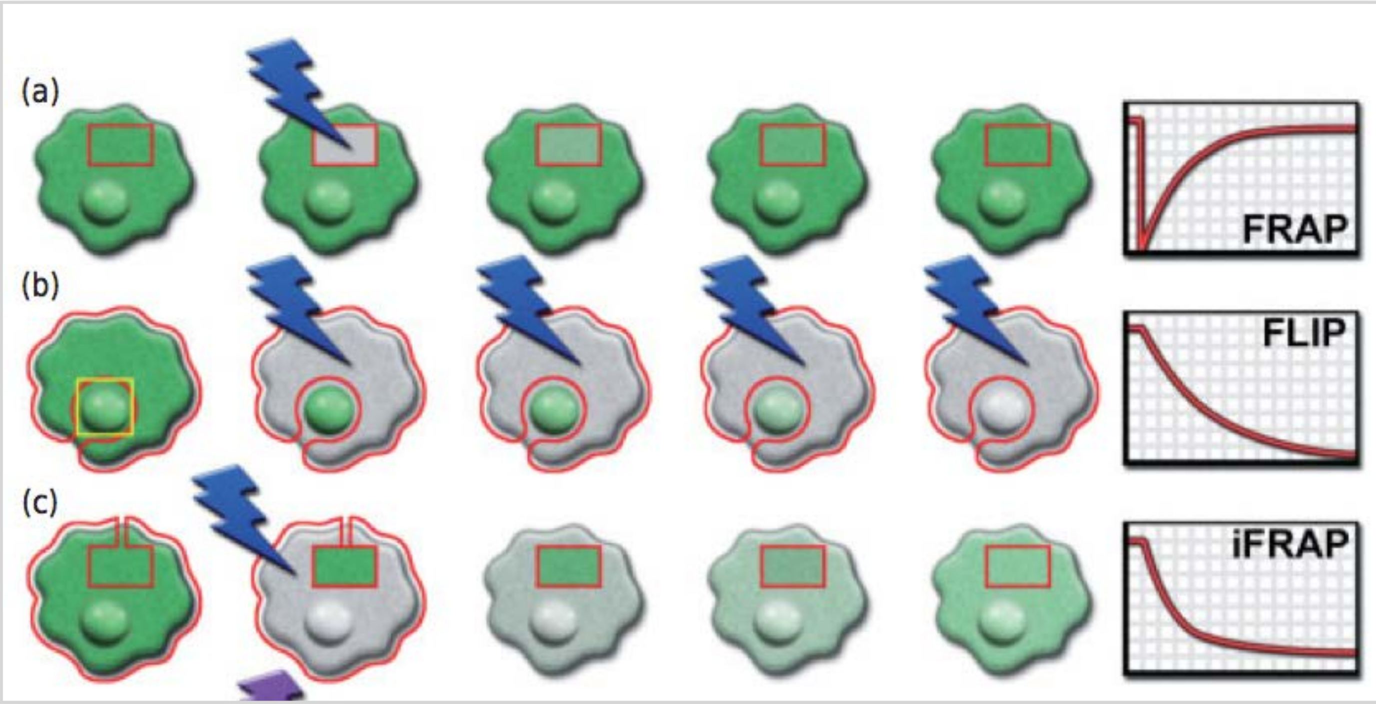
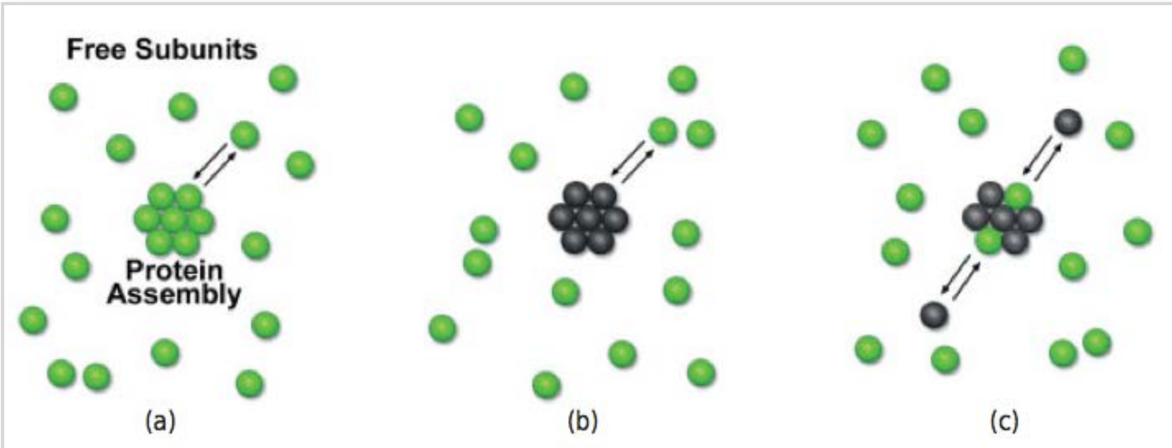
FRAP: Techniques used and related imaging modalities



A molecular model for analysing FRAP measurements



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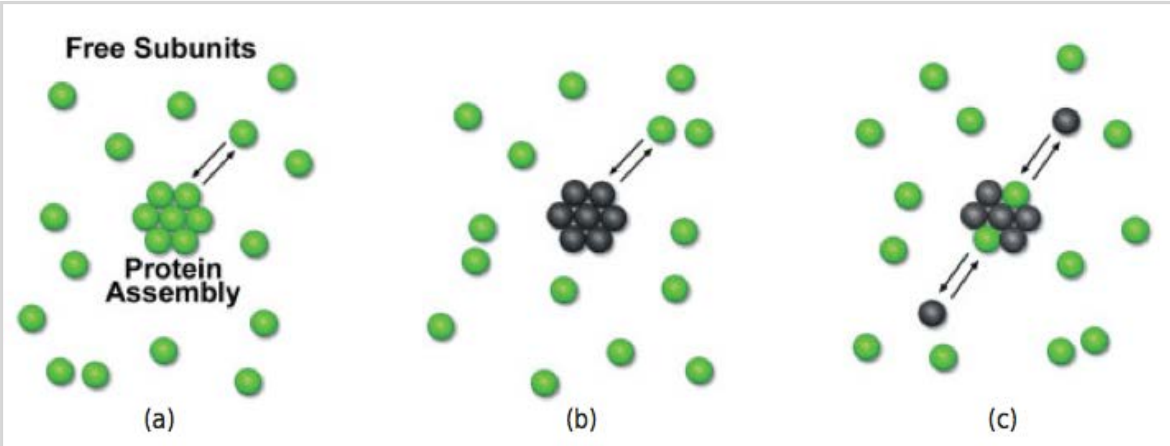
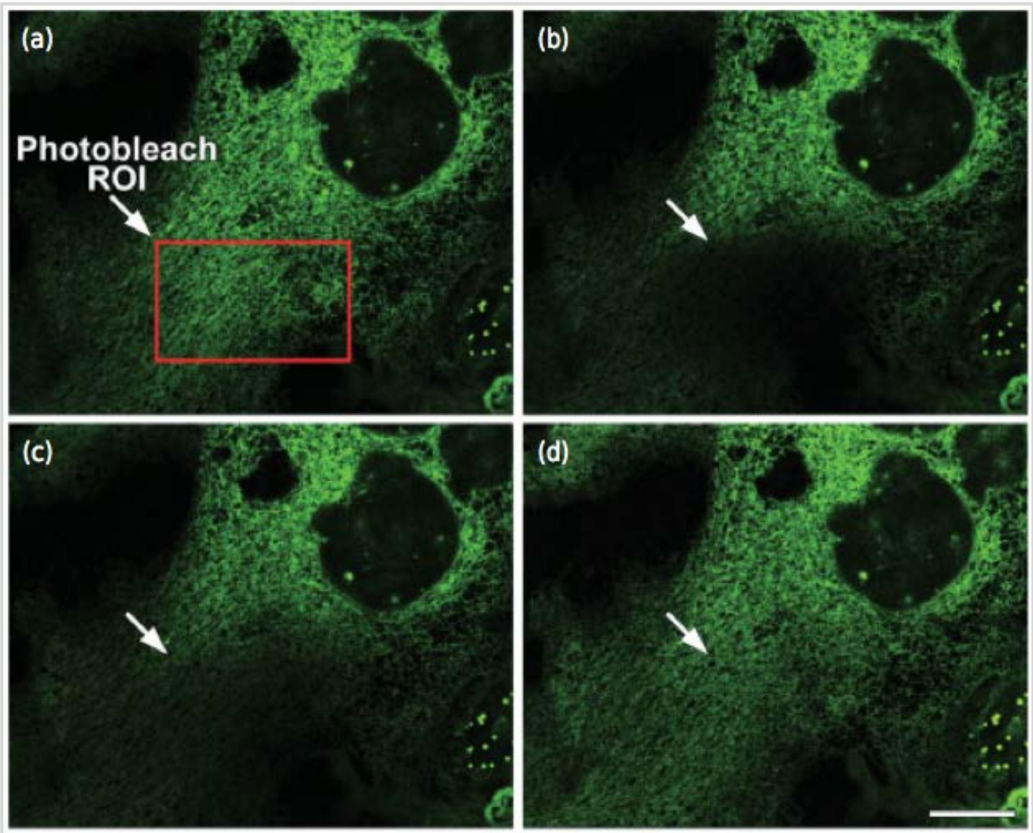


the initial phase of fluorescence recovery is a mixture of two overlapping phases:

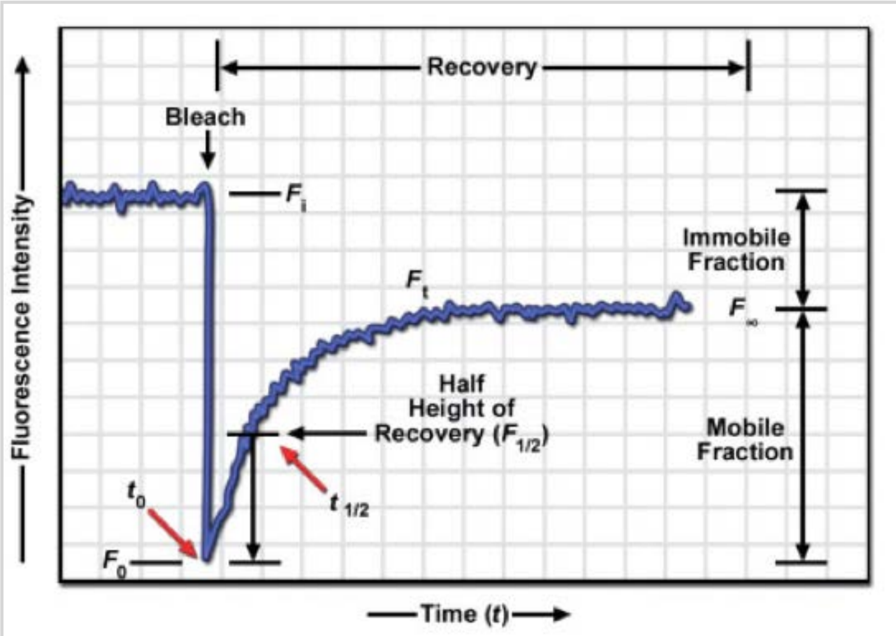
(1) a rapid recovery phase involving diffusion of free fluorescent molecules, where bleached soluble molecules are replaced with fluorescent molecules from outside the bleach zone

(2) a slower overlapping phase based on *recovery of the bleached structure* that depends on the replacement of bleached molecules with new fluorescent ones.

A FRAP experiment



FRAP kinetics



The **mobile fraction**, the fraction of molecules that does show recovery, is calculated from the curve as:

(F_∞/F_i),

the balance is the **immobile fraction**:

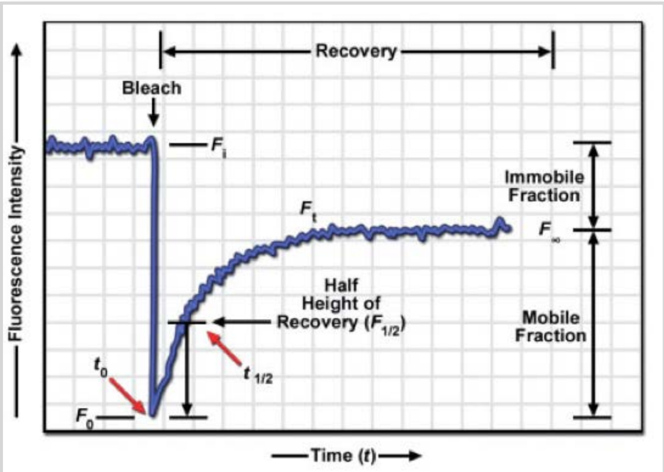
($F_i - F_\infty$)/F_i.

The **percent recovery**, the percent of the initial fluorescence that is recovered, is:

%Recovery = ($F_\infty - F_0$)/($F_i - F_0$),

and F_0 is the fluorescence intensity immediately after bleaching.

FRAP kinetics



The principal goal of FRAP is to determine the characteristic half-time for recovery, t_{half}, from time lapse sequences of the fluorescence recovery.

For simple first-order reactions, a basic approach is to fit FRAP recovery data to the exponential equation:

$$F_t = F_\infty - (F_\infty - F_i)\, e^{-kt},$$

where F_t, F_i, and F_∞ are the fluorescence at time t during recovery, the initial fluorescence, and the final fluorescence, and k is a constant.

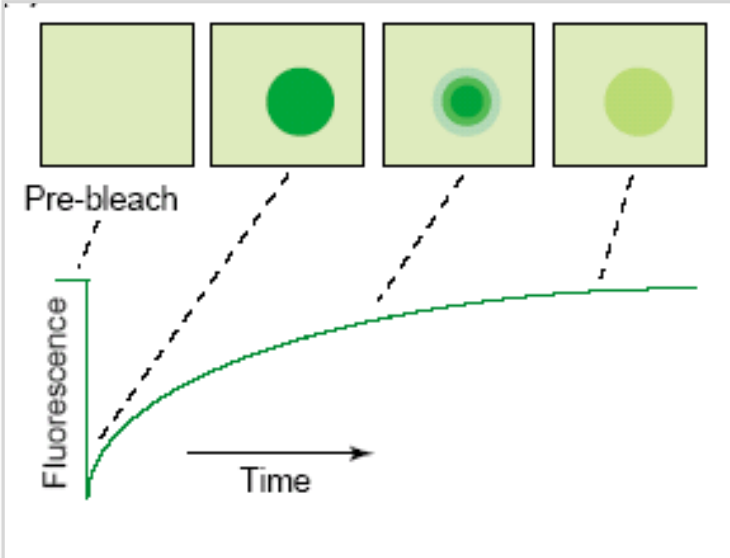
Plotting the natural logarithm of the fluorescence increment $(F_\infty - F_i)$ at each recovery time point versus time:

$$\ln(F_\infty - F_t) = -kt + b.$$

This function gives a straight-line plot, where the slope ($-k$) is related to the recovery half-time and the y-intercept (b) is related to the amount of fluorescence immediately after photobleach.

Measurement of diffusion coefficient D using FRAP

Bleach creates “hole” of fluorophores,
Diffusion is measured by “hole filling in”



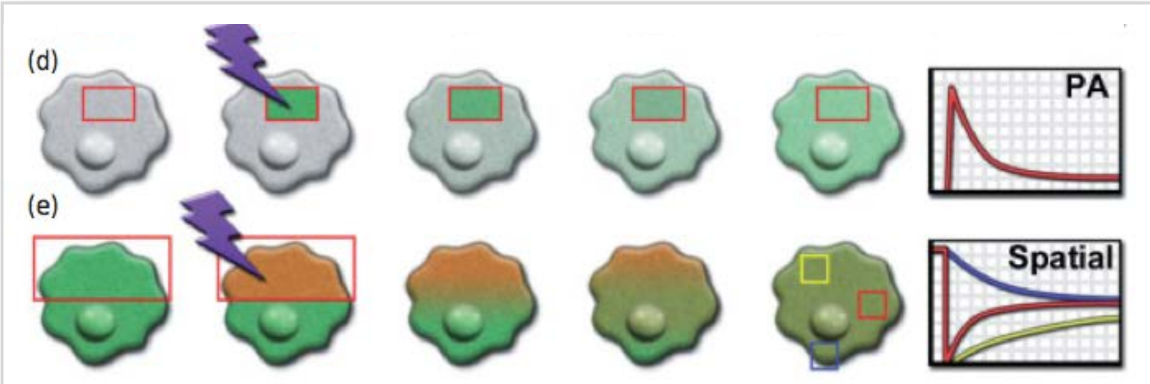
The kinetics of fluorescence recovery of the photobleached zone can then be analysed by the following equation developed by Axelrod et al. (1976) to calculate ***D*** in a two-dimensional system:

$$\tau_D = \omega\gamma/4D,$$

ω = radius of the waist or focused laser beam
γ = photobleaching correction factor
τ_D = time for fluorescence recovery.

Question: What FRAP controls would you use?

FDAP :
Fluorescence decay after photoactivation/-conversion



Question: What is the advantage of FDAP over FRAP?

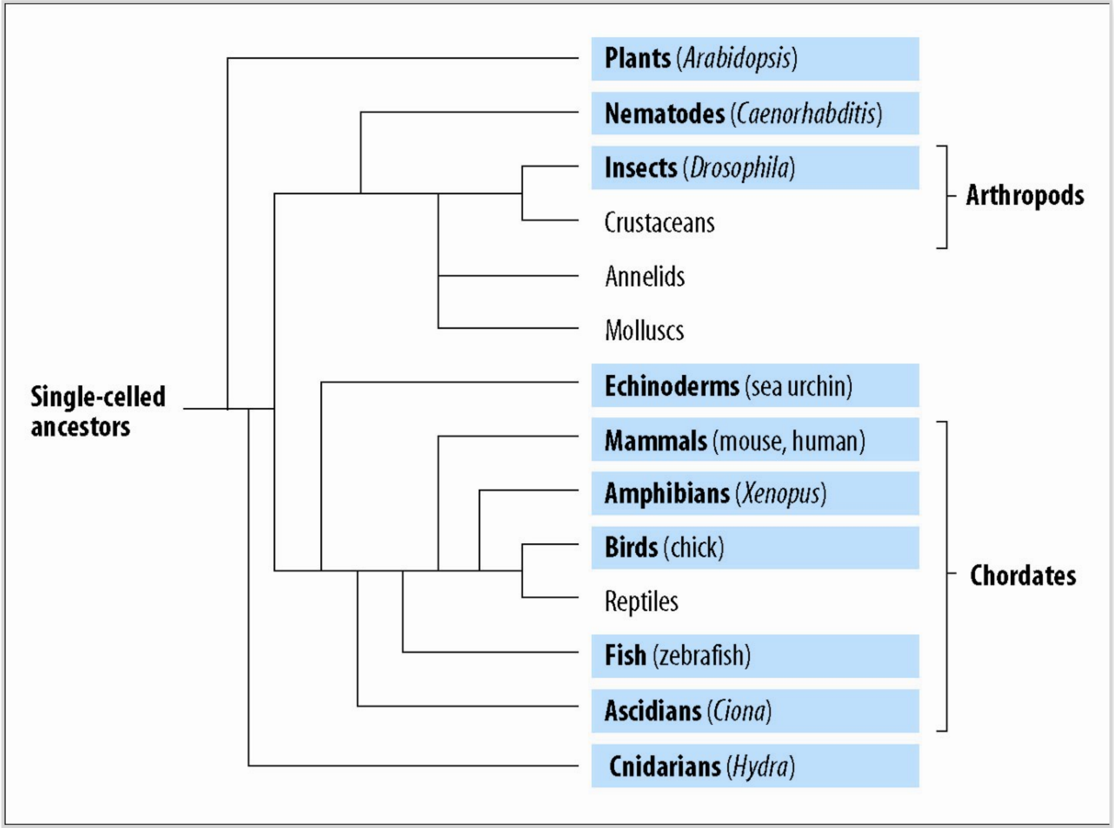
Group discussion: comparison of parameters obtainable with FRAP or FRET?

	FRAP	FRET
Diffusion Rates		
Multicomponent Diffusion		
Mobile fraction		
concentration		
complexing		
binding kinetics		

What we will cover

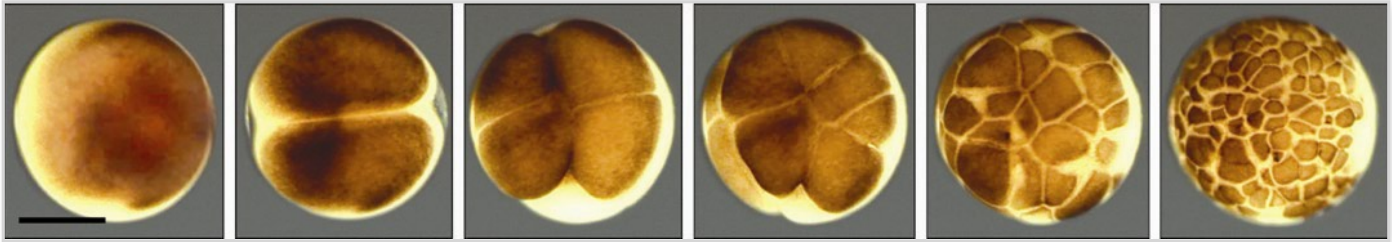
- How to visualise molecular dynamics using fluorescence imaging?
- **What are the basic principles of development?**
- How is tissue patterned?

Development is studied mainly through selected model organisms

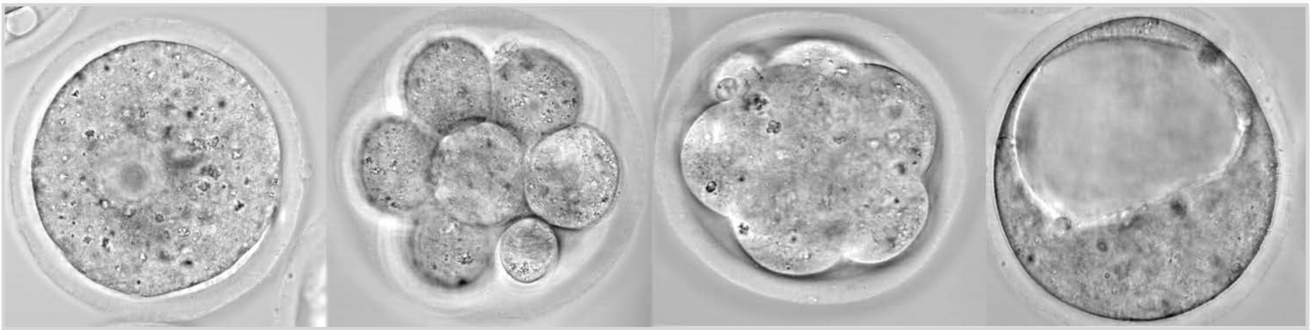


In most embryos, fertilisation follows cell division: the cleavage stage

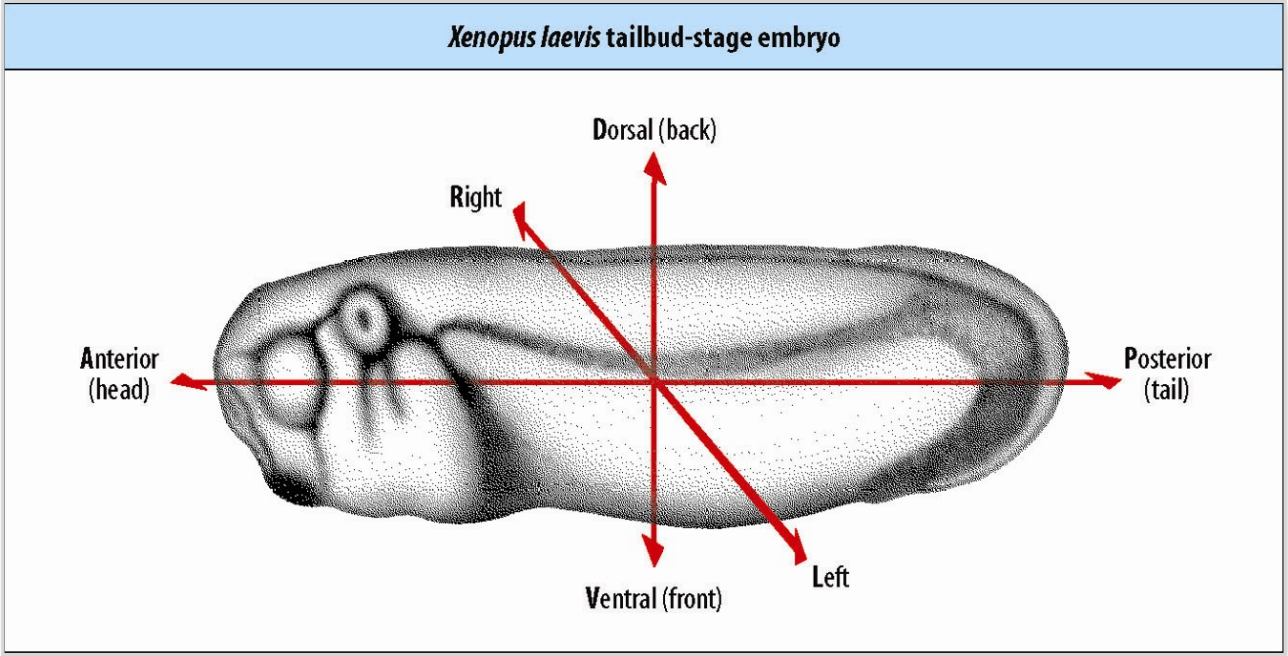
xenopus egg



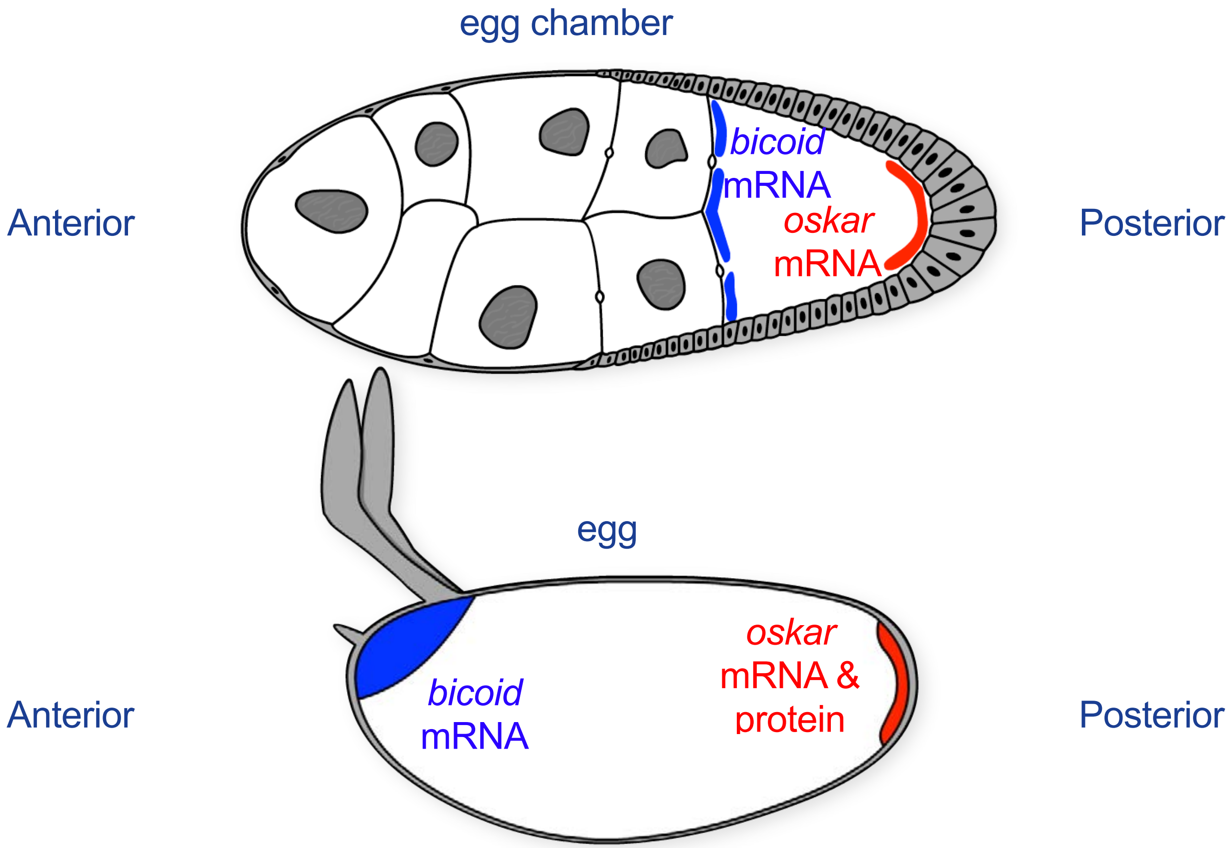
mouse embryo



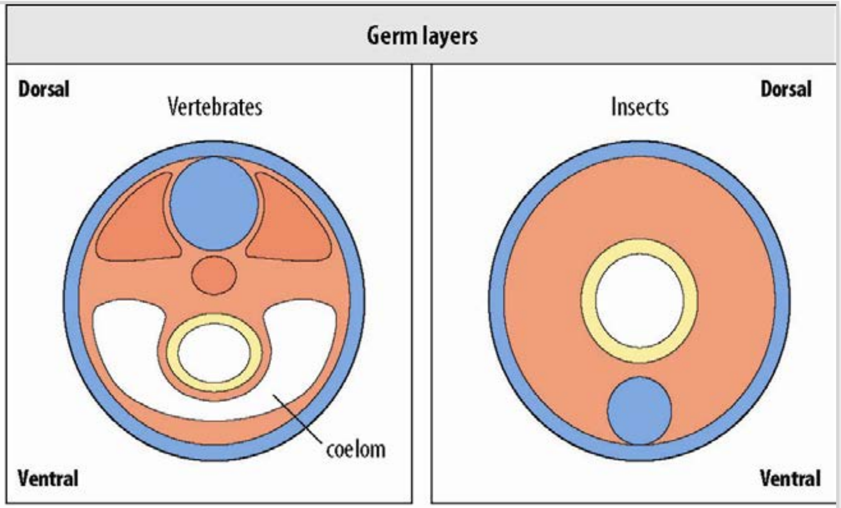
Development involves pattern formation, morphogenesis, cell differentiation and growth



Drosophila melanogaster (fruit fly): polarity laid out during oogenesis

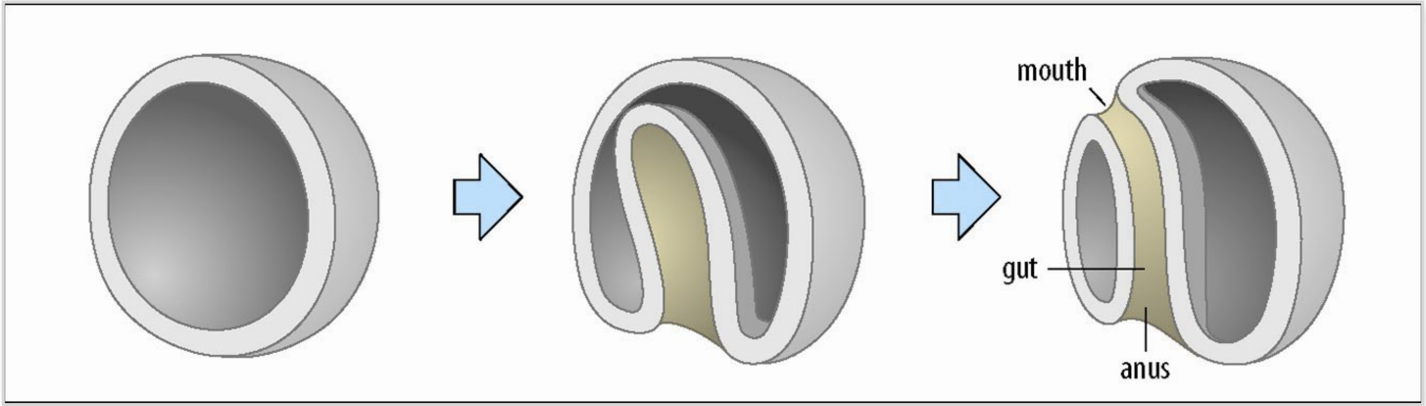


At the same time that the axes are being formed, cells are allocated to different germ layers



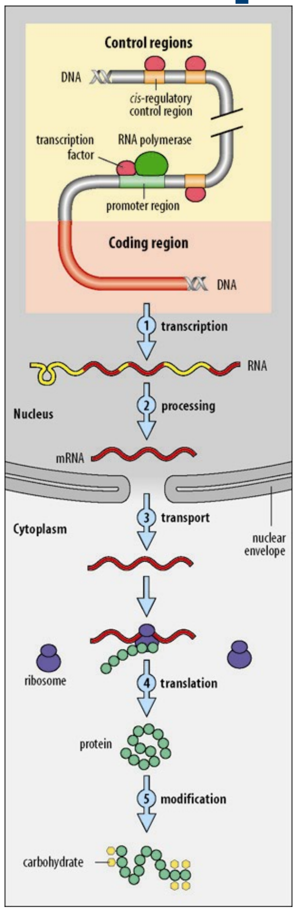
Germ layers	Organs	
Endoderm	gut, liver, lungs	gut
Mesoderm	skeleton, muscle, kidney, heart, blood	muscle, heart, blood
Ectoderm	epidermis of skin, nervous system	cuticle, nervous system

Gastrulation in the sea urchin: dramatic changes in form termed morphogenesis

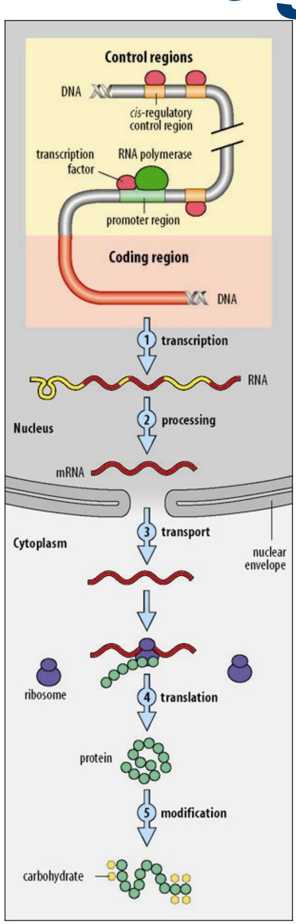


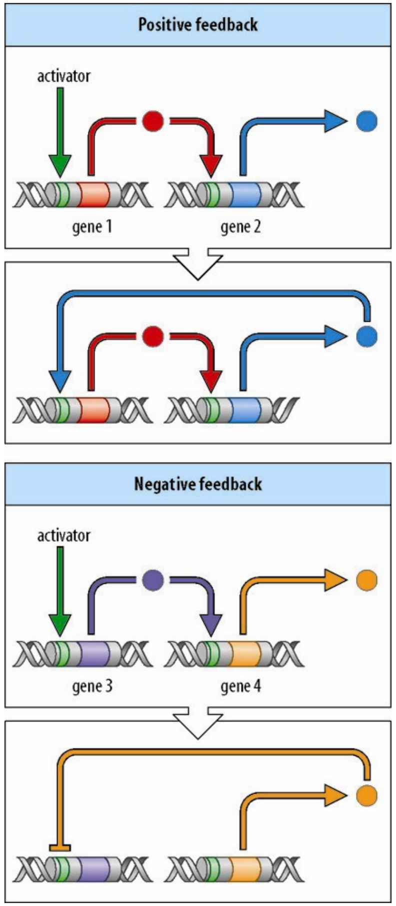
followed by cell differentiation & growth

Genes control cell behaviour by specifying which proteins are made

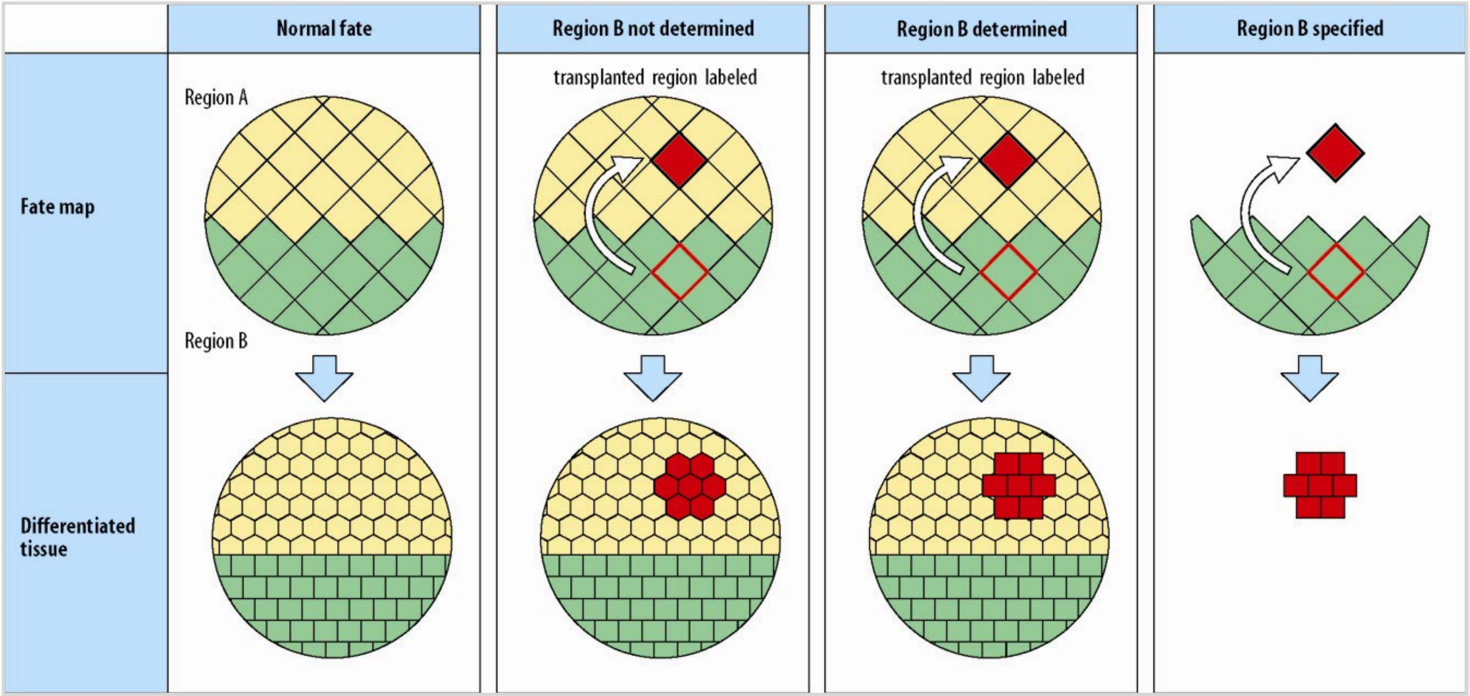


The expression of developmental genes is under tight control





Development is progressive and the fate of cells becomes determined at different times



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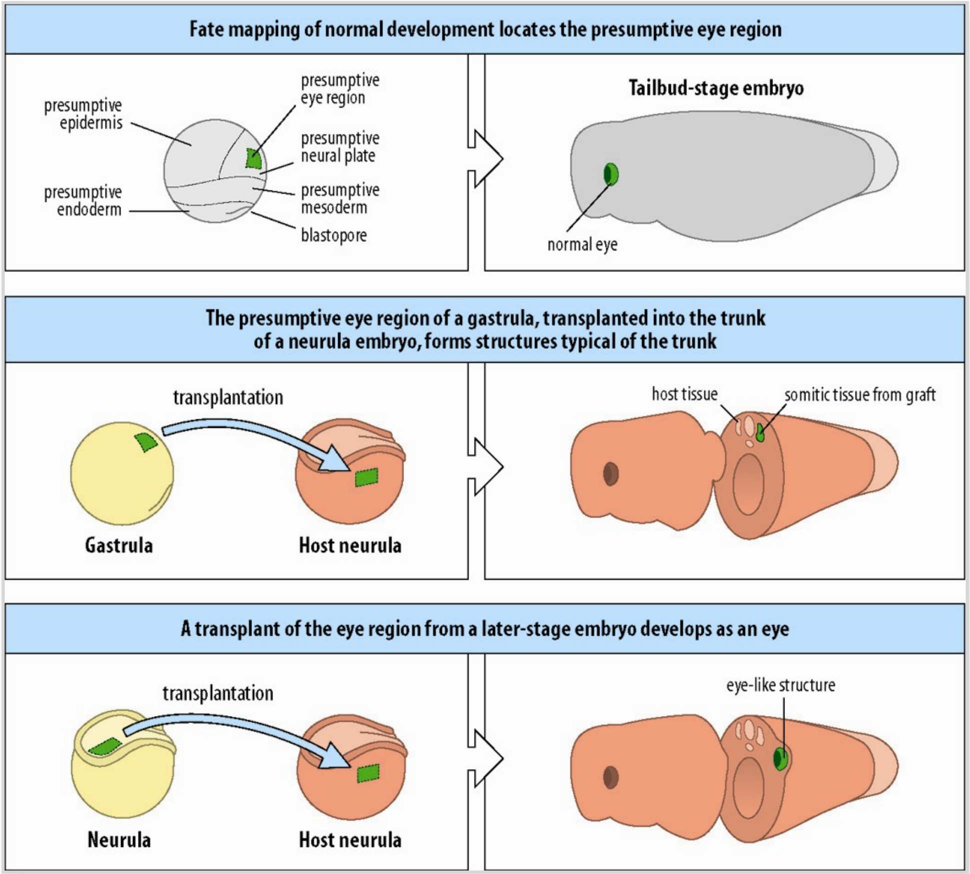
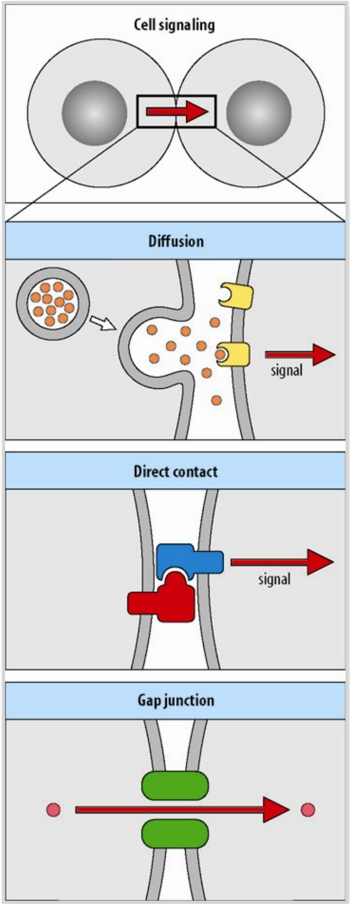


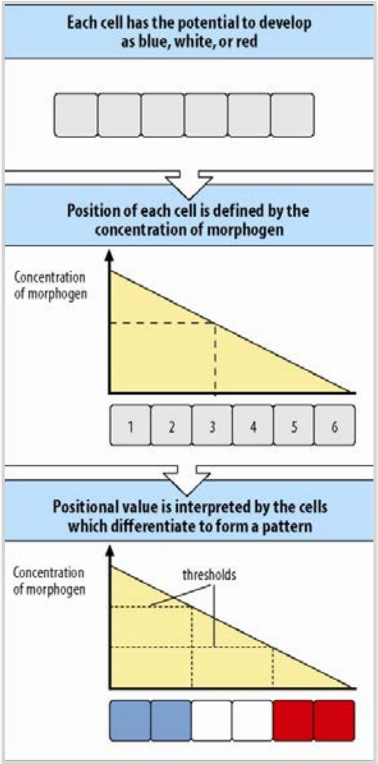
Inductive interactions can make cells different from each other

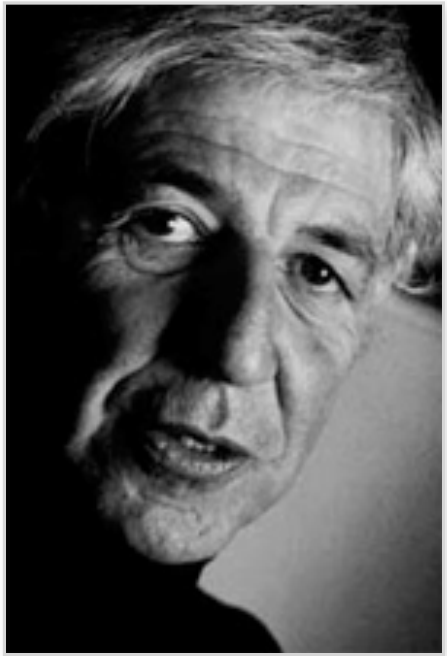


permissive induction:
- one kind of response
- depends on signal threshold

instructive induction:
- different responses to different concentrations

Patterning can involve the interpretation of positional information





Lewis Wolpert (b.1929)

1) positional information has to be related to some boundary
2) this positional information has then to be interpreted
(Note: no need to set relation between positional values and how they are interpreted)

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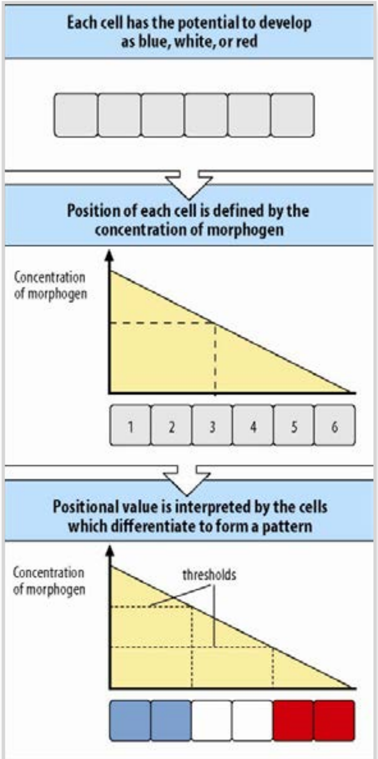

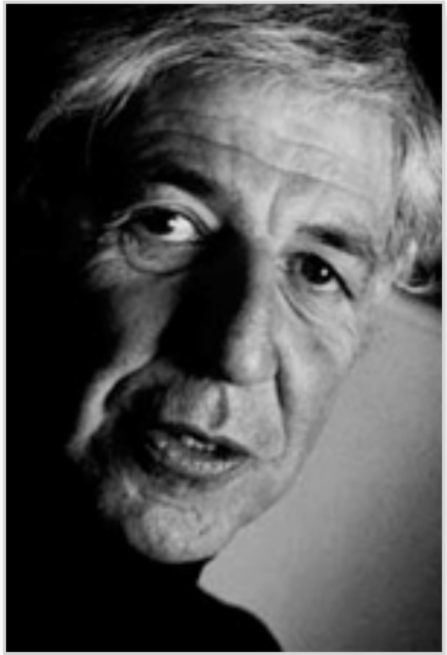

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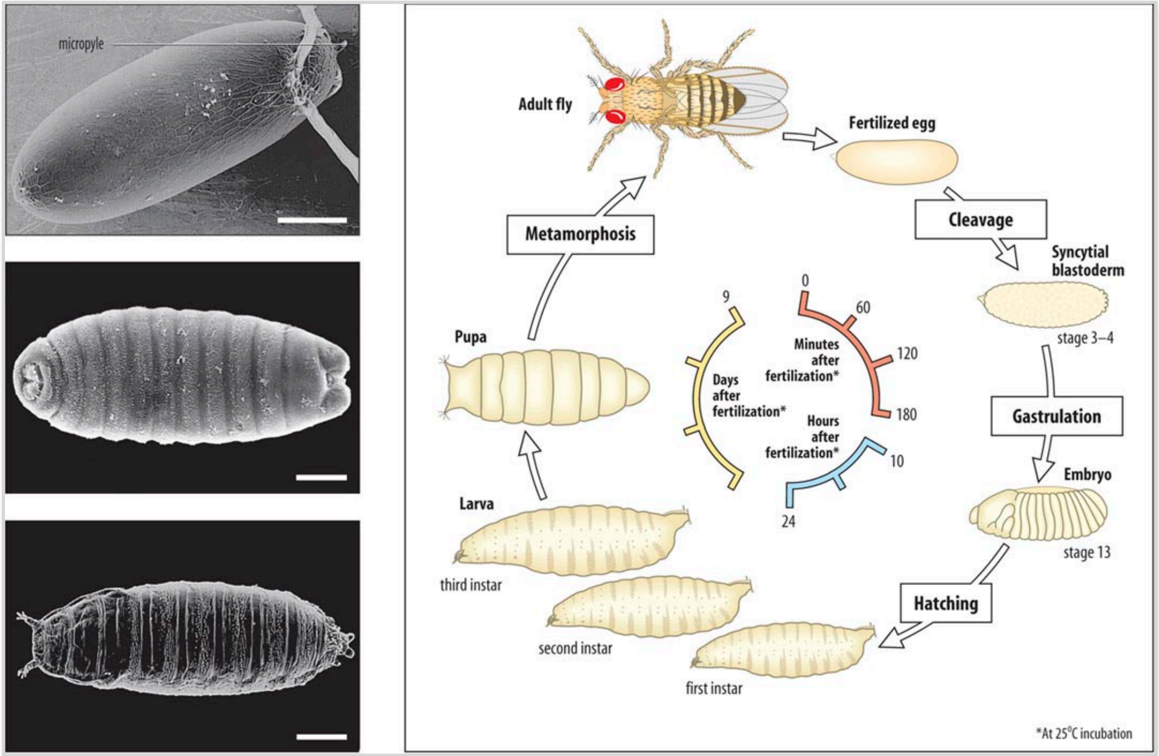
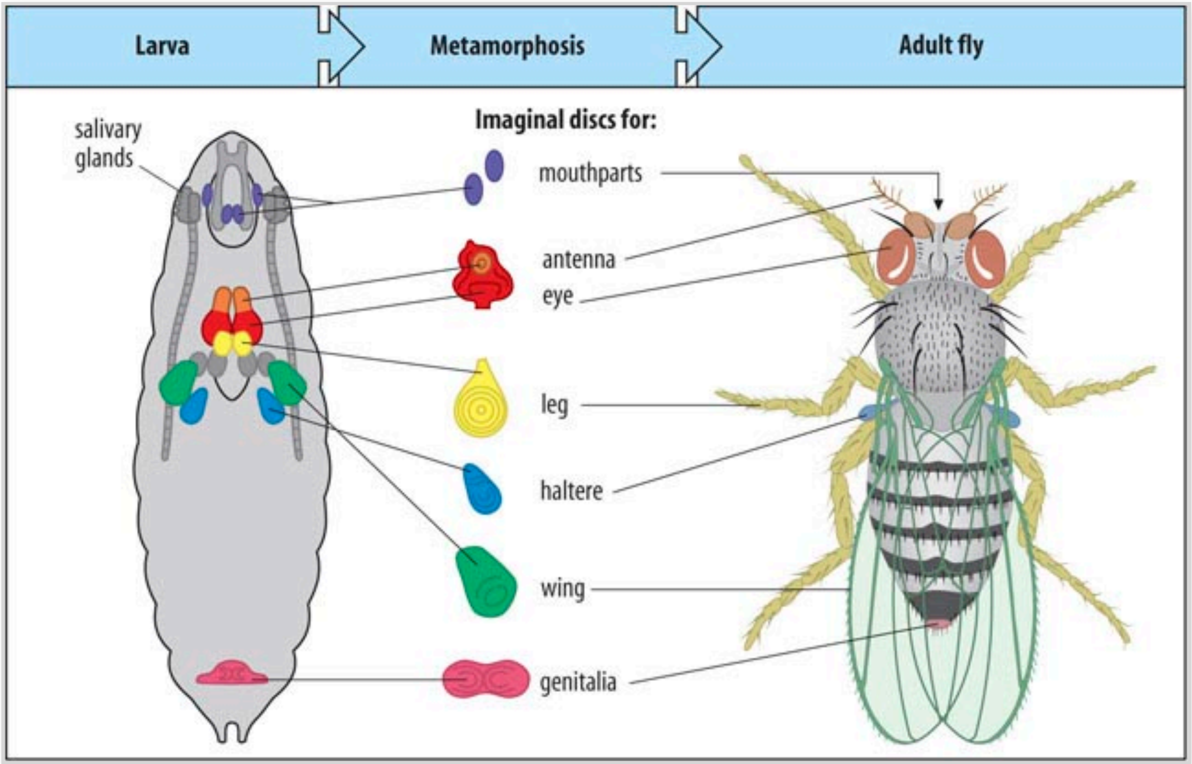
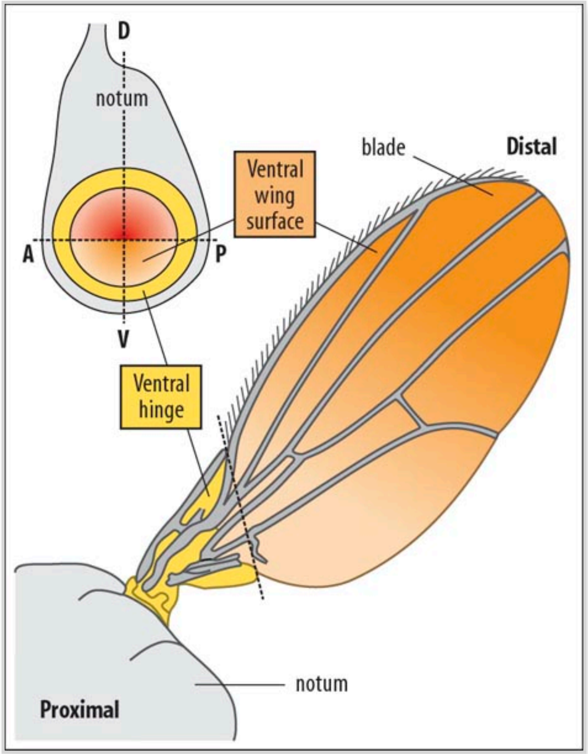
Drosophila melanogaster life cycle and overall development



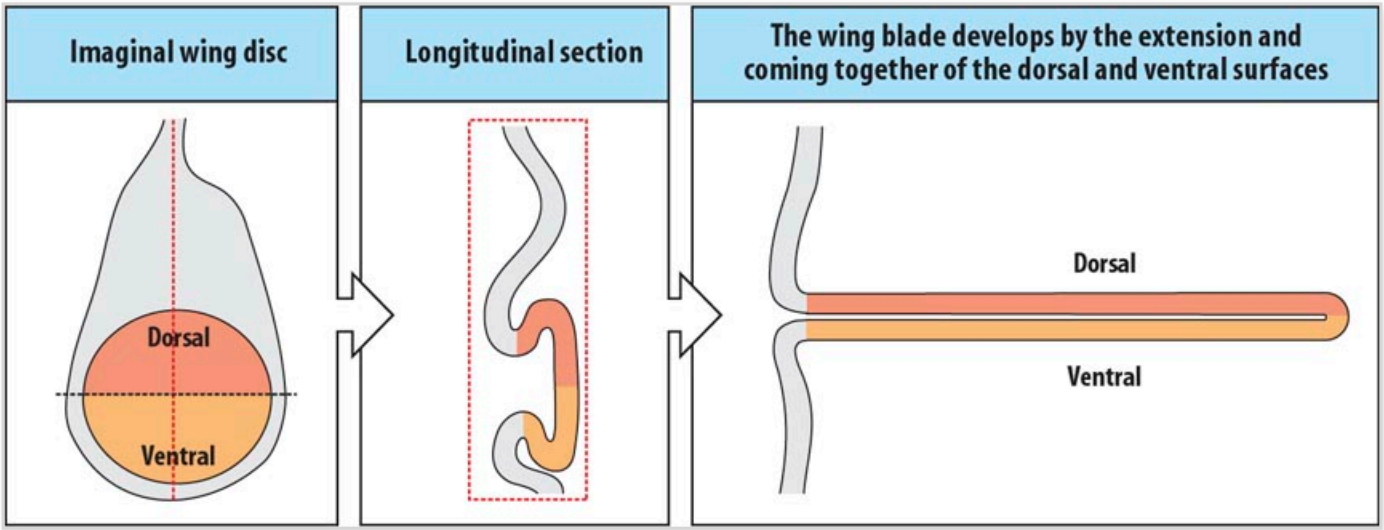
Imaginal discs give rise to adult structures at metamorphosis



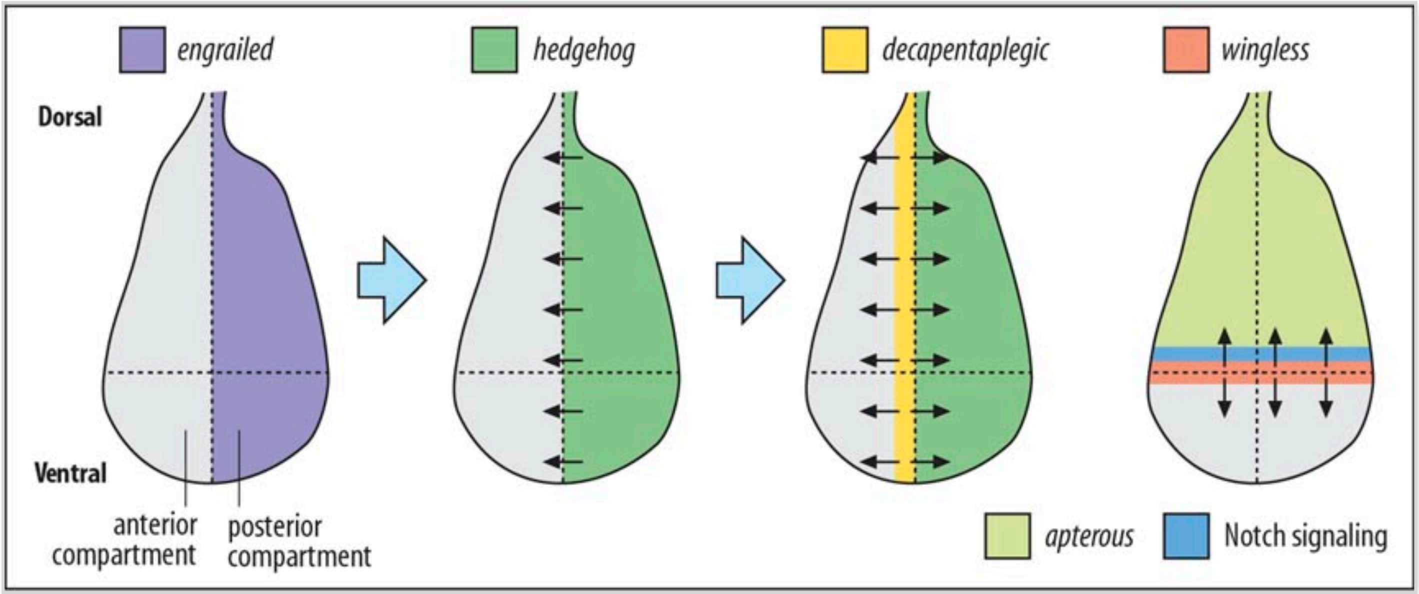
Fate map of the wing imaginal disc of drosophila melanogaster



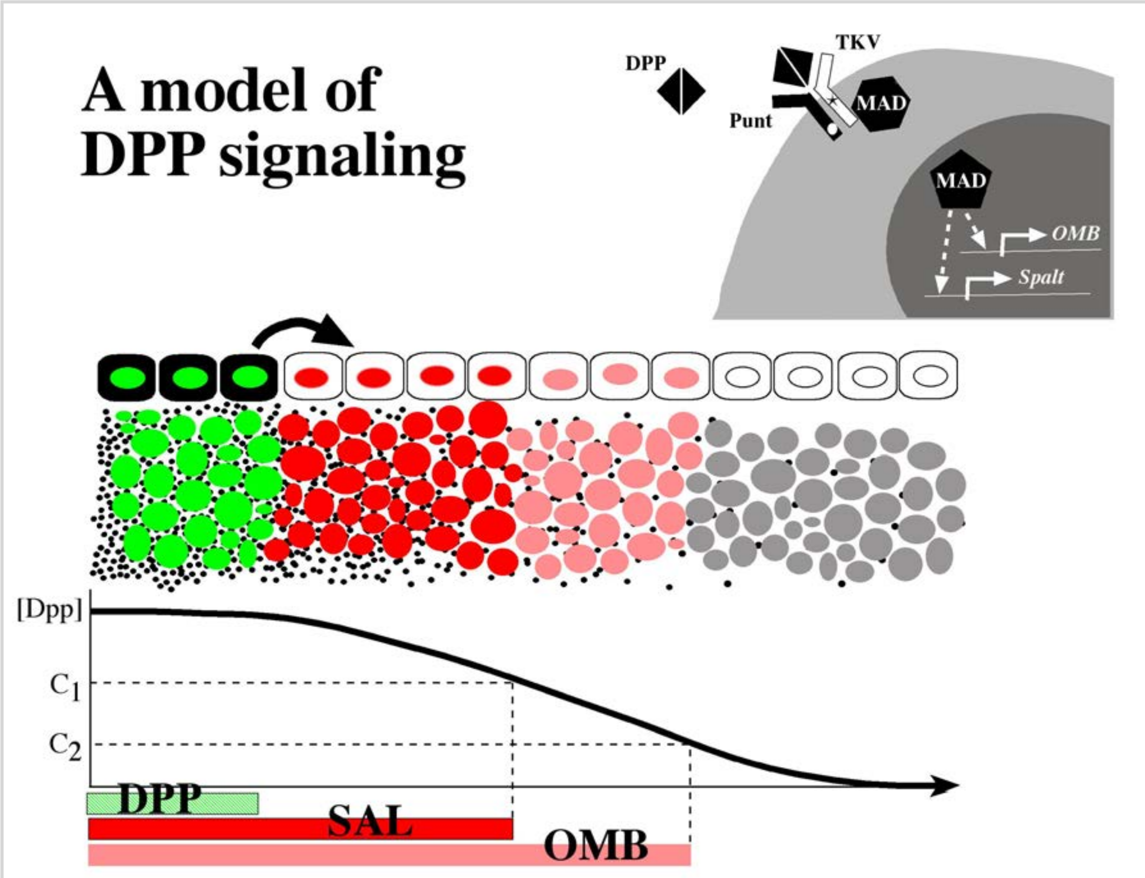
Development of the wing blade from the imaginal disc



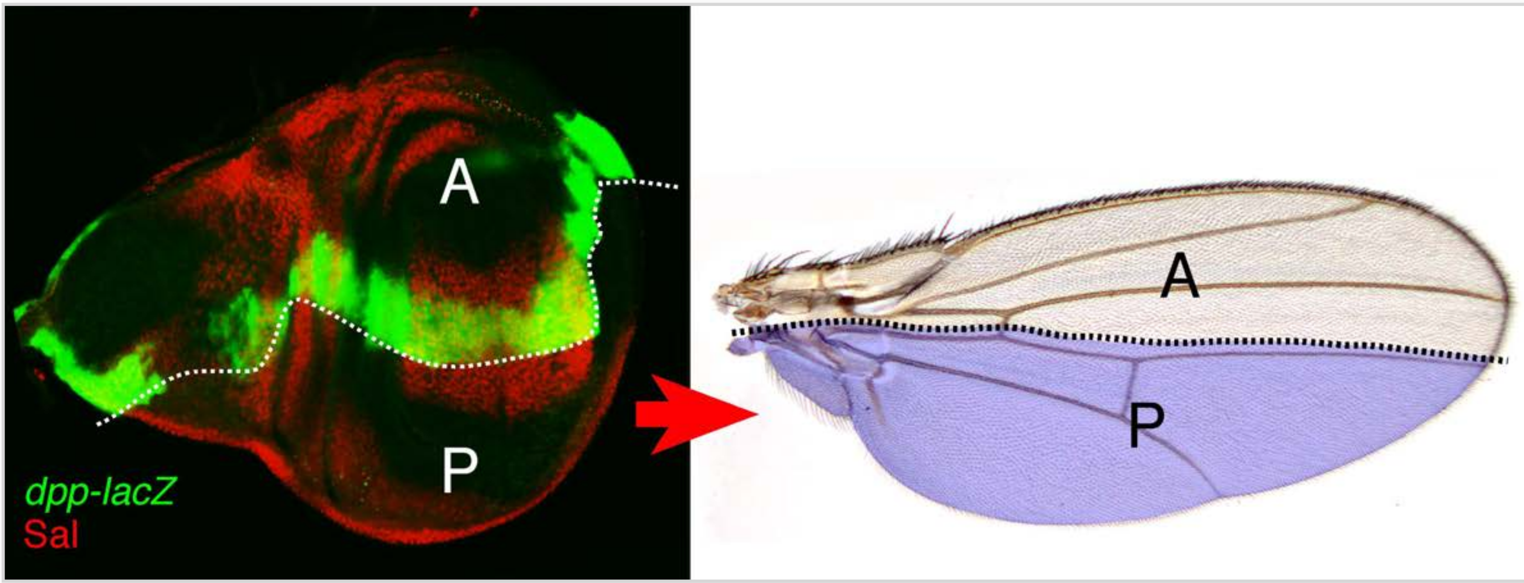
Establishment of signalling regions in the wing disc at the compartment boundaries



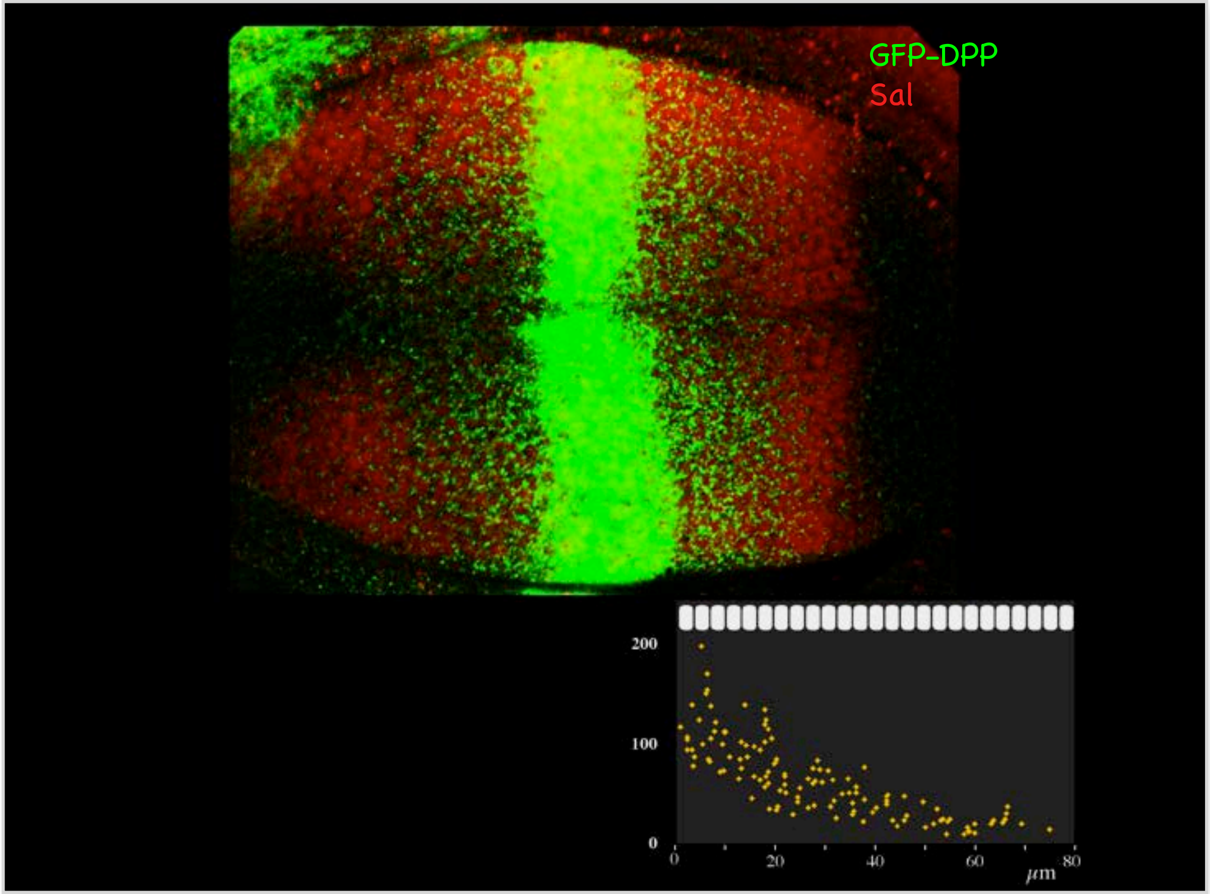
Dpp is a long range morphogen



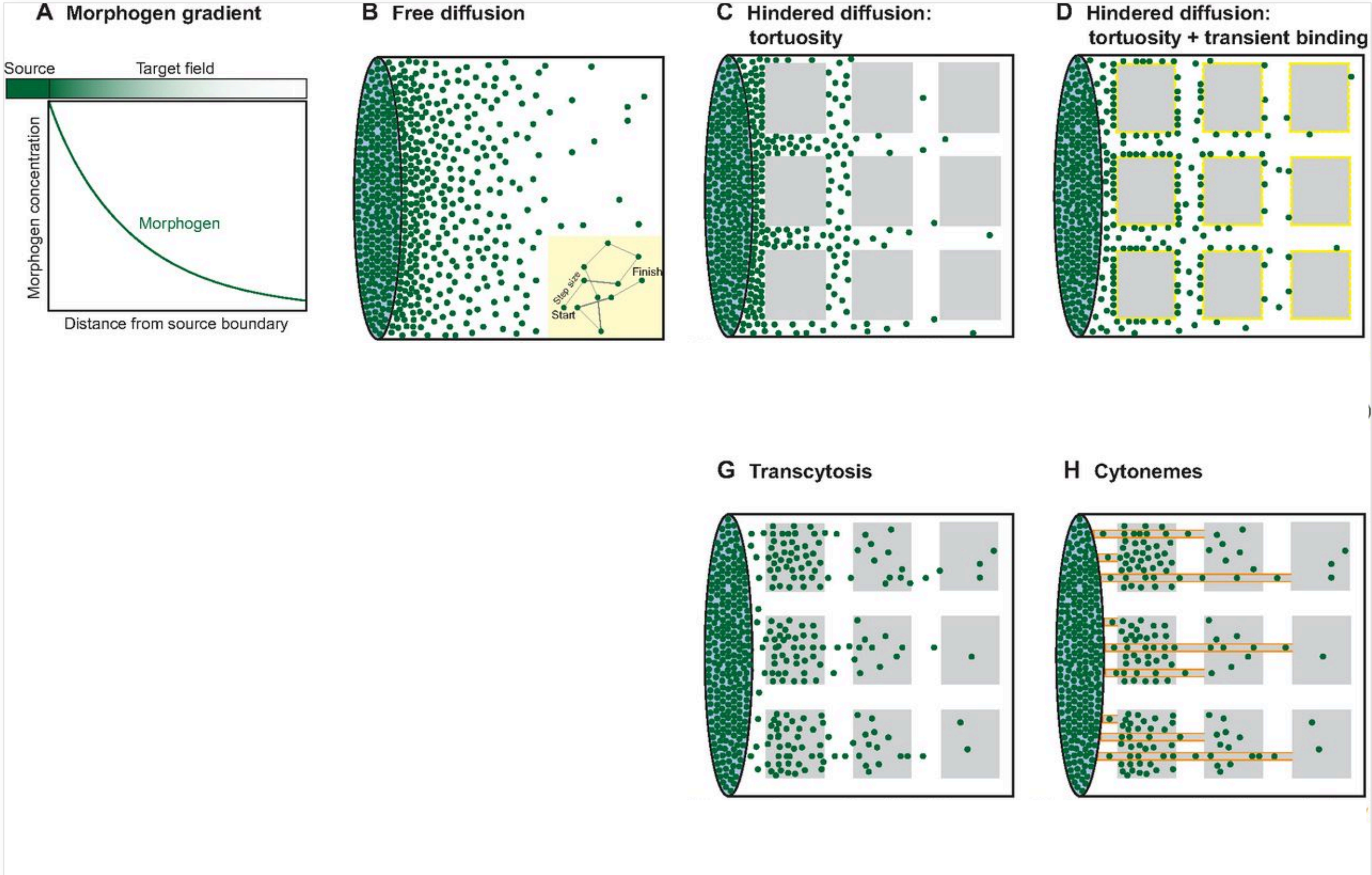
DPP signalling in the wing



DPP is secreted and distributed as a gradient



Morphogen transport models



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Thank you!

Periklis (Laki) Pantazis